

Week 9

Deep Learning

Question: 1

You are building a sentiment analysis model that takes movie reviews as input and classifies them as positive or negative. You choose to represent each word using one-hot encoding. You represent each word in your vocabulary using one-hot encoding. Your vocabulary has 10,000 unique words. How many non-zero elements will a review of 15 words have in its vector representation if you concatenate the one-hot vectors?

Options:

- A. 15
- B. 150,000
- C. 10,000
- D. It depends on the number of unique words in the review

Correct Answer: A

Question: 2

Which of the following issues are most likely to affect the model due to the use of one-hot encoding?

Options:

- A. High computational and memory cost
- B. Loss of word ordering
- C. Inability to capture word similarity
- D. Overfitting due to redundant features

Correct Answers: A and C

Explanation: One-hot encoding results in very high-dimensional sparse vectors, leading to high computational cost. Also, it cannot capture semantic similarity between words like “good” and “excellent”.

Data for the following two questions (3 and 4)

Suppose your vocabulary consists of only 7 words: *cat*, *dog*, *truck*, *car*, *apple*, *banana*, *lion*. Each is represented using one-hot vectors of size 7.

Question: 3

What is the cosine similarity between cat and dog in this one-hot setup?

Options:

- A. 0
- B. 1
- C. $\sqrt{2}$
- D. Undefined

Correct Answer: A

Explanation: One-hot vectors are orthogonal. Therefore, the dot product is 0, and since their magnitudes are 1, cosine similarity = 0.

Question: 4

In the same setup, what is the Euclidean distance between any two distinct one-hot vectors?

Options:

- A. 0
- B. 1
- C. $\sqrt{2}$
- D. 2

Answer: C

Explanation: Two one-hot vectors differ in exactly two positions: one is 1 and the other is 0 in both directions. Hence, Euclidean distance is $\sqrt{((1 - 0)^2 + (0 - 1)^2)} = \sqrt{2}$.

Question: 5

You are analyzing the impact of vocabulary size on model design. You are considering two corpora with vocabularies of size 10,000 and 1,000,000 respectively. What is the primary drawback of using one-hot vectors for the larger vocabulary (1 million words)?

Options:

- A. Increased model bias
- B. Reduced interpretability
- C. Massive increase in dimensionality
- D. Higher risk of synonym confusion

Correct Answer: C

Explanation: One-hot encoding creates vectors of size equal to the vocabulary. With 1 million words, each word is represented using a 1 million-dimensional vector, which is inefficient.

Question: 6

You are building a sentiment analysis model that uses one-hot encoding to represent individual words. The model misclassifies the sentence:

“The plot was as good as a dog's breakfast.”

as positive, even though the phrase “dog’s breakfast” is commonly used to describe a mess or failure, which carries a negative sentiment. The model fails to interpret this nuance correctly. Which key limitation of one-hot encoding most directly contributes to this misclassification?

Options:

- A. Sparsity
- B. Lack of phrase modeling
- C. No syntactic structure
- D. No semantic understanding

Correct Answer: D

Question: 7

You are designing a neural network to process text data using an embedding layer. The vocabulary consists of 5,000 unique words. Initially, you consider using one-hot vectors directly as input.

Which of the following is not a disadvantage of using one-hot vectors directly as input to the model?

Options:

- A. Fixed vocabulary size
- B. Large memory requirements
- C. Inability to capture word similarity
- D. Easy to train with gradient descent

Correct Answer: D

Explanation: One-hot vectors are:

High-dimensional and sparse

Unable to encode relationships between words (like similarity or context)

Not learnable representations — they are fixed and do not update via training

Question: 8

You are working with a vocabulary of size $V = 10,000$. You want to represent a document that contains 50 words by concatenating the one-hot vectors of each word into a single long vector.

What is the sparsity percentage of the document vector described above?

$$\text{Sparsity} = \left(\frac{\text{Number of zero entries}}{\text{Total number of entries}} \right) \times 100$$

Options:

- A. 99.09%

- B. 99.99%
- C. 99.0%
- D. 95%

Correct Answer : B

Explanation:

$$Sparsity = \left(\frac{499,950}{500,000} \right) \times 100 = 99.99\%$$

Question: 9

In your corpus, common words like "*the*", "*is*", and "*and*" appear very frequently and dominate co-occurrence counts, making it harder for your model to capture meaningful semantic relationships between content words.

Which of the following are valid techniques to address the over-representation of high-frequency words in co-occurrence matrices?

Options:

- A. Drop high-frequency words from the vocabulary
- B. Apply a frequency threshold to clip their co-occurrence counts
- C. Set their PMI to zero for all co-occurrences
- D. Assign them higher weights during training to emphasize their influence

Correct Answers: A, B

Question: 10

Which of the following formulas correctly represents frequency thresholding in co-occurrence matrices?

Options:

- A. $PMI(w, c) = 0$ if $count(w, c) = 0$
- B. $X_{ij} = \min(count(w_i, c_j), t)$
- C. $X_{ij} = \max(count(w_i, c_j), t)$
- D. $X_{ij} = count(w_i, c_j)^2$

Correct Answer: B

Question: 11

In your corpus:

- The word w and the context word c co-occur 50 times
- The total number of word-context pairs is 10,000
- The word w appears 200 times
- The context word c appears 500 times

Compute $PMI(w, c)$ using log base 2 and round your answer to two decimal places.

Options:

- A. 1.32
- B. 2.32
- C. 3.32
- D. 4.32

Correct Answer: B

Explanation:

$$PMI(w, c) = \log_2 \left(\frac{50 \times 10000}{200 \times 500} \right) = \log_2 5 \approx 2.32$$

Question: 12

You are working with the sentence:

"The human uses the machine for applications."

You use a context window size of 2, meaning for any target word, you consider up to 2 words to the left and 2 words to the right as its context.

Your target word is: *machine*

How many times does the word “human” appear in the context window of the word “machine”?

Options:

- A. 0
- B. 1
- C. 2
- D. Cannot be determined

Correct Answer: A

Explanation: The word “machine” appears once in the sentence. Its context window includes: 2 words to the left: "uses", "the", 2 words to the right: "for", "applications"

The word “human” is 3 positions before “machine”, so it is outside the context window. Therefore, it is not included as a context word for “machine.”

Question: 13

You are analyzing co-occurrence data for the word “system” with various context words. The raw co-occurrence counts are:

Context Word	Count
response	300
time	120
user	80
ror	500

To avoid overemphasis on high-frequency pairs, you apply frequency thresholding using a threshold value of $t = 100$.

What are the adjusted co-occurrence counts for the word “system” with the given context words after thresholding?

Options:

- A. system–response: 100
- B. system–time: 100
- C. system–user: 80
- D. system–for: 100

Correct Answers: A, B, C, D

Explanation: After applying thresholding :

system–response: $300 \rightarrow 100$ (clipped)

system–time: $120 \rightarrow 100$ (clipped)

system–user: $80 \rightarrow$ remains 80

system–for: $500 \rightarrow 100$ (clipped)

Thus, all options reflect the correct adjusted counts.

Question: 14

You start with a co-occurrence matrix of size 8000×8000 (word $\times$ context). You apply truncated Singular Value Decomposition (SVD) to reduce the dimensionality of this matrix to a lower rank $k = 300$, in order to obtain dense word vectors for downstream tasks.

What is the size of the resulting word representation matrix after dimensionality reduction?

Options:

- A. 8000×8000
- B. 8000×300
- C. 300×8000
- D. 300×300

Correct Answer: B

Question: 15

Which statements about using SVD for word similarity are true?

Options:

- A. Words with similar contexts are close in vector space
- B. Orthogonality is preserved in $U \Sigma$
- C. Word analogies are captured better with reduced vectors
- D. One-hot encodings are better for similarity than $U \Sigma$

Correct Answers: A, C

Explanation: SVD encodes semantic similarity and can even capture analogical structure in reduced space.

Question: 16

You cluster the low-dimensional word vectors obtained using $U \Sigma$. Words like “cat”, “dog”, “tiger” cluster together, while “plane”, “car” form another group. What property of SVD enables such clustering?

Options:

- A. Orthogonal context vectors
- B. Co-occurrence counting
- C. Capturing latent semantics via singular vectors
- D. TF-IDF-based filtering

Correct Answer: C

Explanation: SVD uncovers hidden structure in word usage, enabling grouping by meaning.

Question: 17

You have two low-dimensional word vectors obtained using $U_k \Sigma_k$
 $w_1 = [0.6, 0.8], w_2 = [0.9, 0.1]$

What is the cosine similarity between w_1 and w_2 ?

Options:

- A. 0.69
- B. 0.81
- C. 0.55
- D. 0.90

Correct Answer: A

Explanation:

$$\begin{aligned} w_1 \cdot w_2 &= (0.6)(0.9) + (0.8)(0.1) = 0.54 + 0.08 = 0.62 \\ \|w_1\| &= \sqrt{0.6^2 + 0.8^2} = 1 \\ \|w_2\| &= \sqrt{0.9^2 + 0.1^2} \approx 0.905 \\ \cos(\theta) &= \frac{0.62}{1 \times 0.905} \approx 0.69 \end{aligned}$$

Data for the following three questions (18 to 19)

You are building a chatbot for a banking application. During the development, you want to embed the following key terms:

loan, interest, deposit, withdrawal, and bank.

You try using one-hot vectors, but the chatbot performs poorly when generalizing across semantically similar queries.

Question: 18

What is the primary reason one-hot vectors failed in this task?

Options:

- A. They include stop words
- B. They do not capture semantic similarity between words
- C. They are computationally intensive to compute
- D. They overfit to rare words

Answer: B

Explanation: One-hot vectors are orthogonal; they do not encode closeness between semantically related words like loan and interest.

Question: 19

You then build a co-occurrence matrix using a large banking corpus. What kind of matrix do you obtain?

Options:

- A. Dense matrix with low dimensionality
- B. Sparse matrix with high dimensionality
- C. Diagonal matrix
- D. Identity matrix

Answer: B

Explanation: Word $\times$ context co-occurrence matrices are typically large and sparse.

Question: 20

To compress the information and obtain semantic vectors, you apply SVD to the matrix. How will this help your chatbot?

Options:

- A. Remove the need for training data
- B. Convert word vectors into binary form
- C. Discover latent topics and reduce noise
- D. Make the matrix symmetric

Correct Answer: C

Explanation: SVD captures hidden semantic structure and produces compact, meaningful word representations.

Question: 21

Which of the following correctly compares one-hot encoding, co-occurrence-based vectors, and SVD-based embeddings?

Options:

- A. One-hot and SVD-based embeddings are both sparse
- B. SVD-based embeddings are lower-dimensional and capture latent meaning
- C. Co-occurrence vectors are dense and outperform embeddings
- D. SVD introduces noise by keeping all singular values

Answer: B

Question: 22

You are training a Word2Vec Skip-gram model on a large corpus containing millions of sentences. During training, you observe that computing the softmax function at the output layer becomes the main bottleneck, especially because the vocabulary size is ~500,000 words. To reduce training time without significantly compromising embedding quality, your team decides to explore alternate loss functions and optimization strategies. Which of the following modifications would best address the computational inefficiency at the output layer while still allowing the model to learn meaningful word embeddings?

Options:

- A) Use a traditional softmax layer and increase the number of hidden units
- B) Switch to negative sampling, training the model to distinguish between real and randomly sampled words
- C) Use one-hot encoding for words and apply cosine similarity for prediction
- D) Remove the output layer entirely and only use context window averaging

Correct Answer: B

Question: 23

You are training a Word2Vec Skip-gram model with negative sampling. For each target-context pair (w, c) , you generate 5 negative samples (w, r) . You sample these negative words r from a modified distribution:

$$p(r) \propto \text{count}(r)^{\frac{3}{4}}$$

Your teammate asks: Why raise the count to the power of $\frac{3}{4}$ instead of using the raw unigram distribution?

What is the main purpose of using $p(r) \propto \text{count}(r)^{\frac{3}{4}}$ for sampling negative words?

Options:

- A) It reduces the dominance of high-frequency but uninformative words like “the”, “and”
- B) It improves the accuracy of top-k predictions in the softmax layer
- C) It smoothes the unigram distribution to prevent overfitting to frequent words
- D) It ensures that all words are sampled equally for fairness

Correct Answers: A, C

Explanation: High-frequency stopwords dominate the raw unigram distribution. Raising to $\frac{3}{4}$ reduces their impact.

The power $\frac{3}{4}$ smoothes the distribution(not too flat and sharp), which helps generalization.

Question: 24

You're training Word2Vec using the following objective for each pair (w, c) and its negative samples (w, r) :

$$L = \log \sigma(v_w^T u_c) + \sum_{r=1}^k \log \sigma(-v_w^T u_r)$$

After a few epochs, you observe that the dot products $v_w^T u$ (for all words) are very small, and all embeddings seem to shrink toward the origin.

What could be valid interpretations or consequences of this observation?

Options:

- A. The dot products approaching zero make sigmoid outputs close to 0.5, resulting in low gradient updates
- B. The model is under-penalizing negative samples due to insufficient contrast between $v_w^T u_c$ and $v_w^T u_r$.
- C. The loss function explicitly encourages dot products for negative pairs to be strongly negative, not just zero
- D. Embeddings collapsing to zero minimize the loss and represent an optimal solution

Correct Answers: A, B, C

Question: 25

Why is negative sampling used in training the Skip-gram model?

Options:

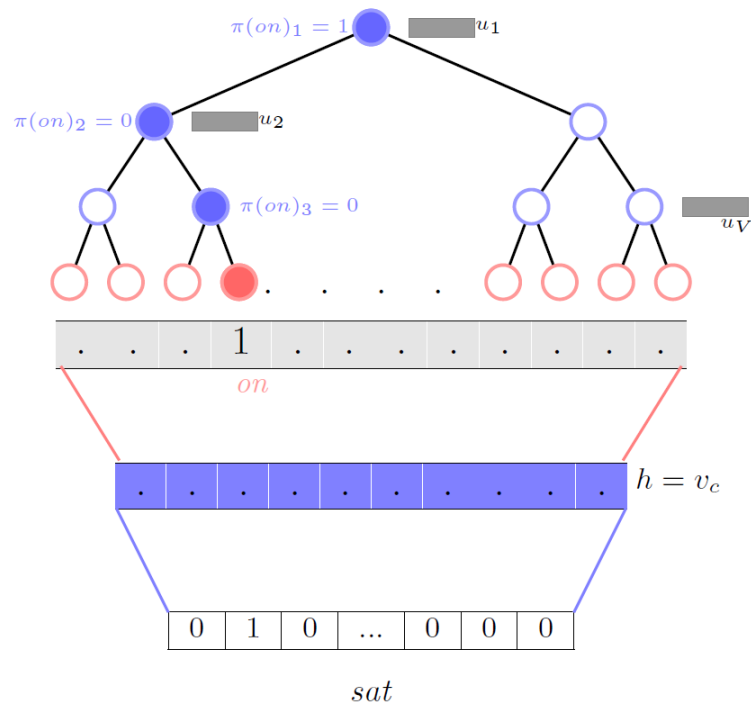
- A. To increase the number of output classes in the softmax layer
- B. To reduce computational cost by avoiding full softmax over large vocabulary
- C. To ensure that all context words are equally likely
- D. To generate multiple embeddings for each word

Correct Answer: B

Explanation: Negative sampling replaces the expensive full softmax over the vocabulary with a lightweight approximation by only updating a few sampled "negative" examples per step.

Question: 26

Suppose the context word is "sat", and the target word is "on". The path $\pi(\text{"on"}) = [1, 0, 0]$.



What does the probability $P(\text{"on"} \mid v_{\text{sat}})$ decompose into in hierarchical softmax?

Options:

- A. A single softmax over all words
- B. A product of three softmax values
- C. A product of sigmoid outputs for each decision on the path
- D. A dot product between v_{sat} and embedding of "on"

Correct Answer: C

Question: 27

In hierarchical softmax, a sigmoid is used at each internal node. Let node u_i appear on the path to word w . The path branches left at u_i .

How is $P(\pi(w)_i = 1)$ computed?

Options:

- A) $\frac{1}{1+e^{-u_i^T v_c}}$
- B) $\frac{1}{1+e^{u_i^T v_c}}$
- C) $1 - \frac{1}{1+e^{u_i^T v_c}}$
- D) $v_c^T u_i$

Correct Answer: A

Question: 28

What are the parameters that must be learned during training while building a hierarchical softmax module?

Options:

- A) Context word vectors v_c
- B) Binary vector $\pi(w)$
- C) Node Vectors $u_1, u_2, \dots u_V$
- D) Output word embeddings directly

Correct Answers: A, C

Question: 29

Given the Term, “*imposed*” with the Vector: $v_{imposed} = [0.5, 0.7, 0.2]$ and few candidate words “enforced, suggested, allowed, and ignored”. The vectors of the candidate words are $v_{enforced} = [0.45, 0.75, 0.25]$, $v_{suggested} = [0.01, 0.9, 0.3]$, $v_{allowed} = [0.8, 0.2, 0.1]$, $v_{ignored} = [0.2, 0.1, 0.9]$.

Which of the following is the most appropriate synonym of the word *imposed*, based on cosine similarity?

Options:

- A) enforced
- B) suggested
- C) allowed
- D) ignored

Correct Answer: A

Question: 30

In Word2Vec, the Skip-Gram model:

Options:

- A. Predicts the center word from its context words
- B. Predicts context words from the center word
- C. Assigns one-hot vectors to words
- D. Minimizes the Hamming distance between vectors

Correct Answer: B